Date: Day:

OS Assignment 02

Samiullah Ashfaq SP24-BSE-107

MLFQ Scheduling Problem

Q_1: Round Robin, $q = 4$ (Highest Priority)
Q_2: Round Robin, $q = 8$
Q_3: First-Come-First-Served (Lowest Priority)

Movement: All processes start in Q_1, if they exhaust a quantum, they are demoted to the next lower queue.

Preemption: A higher-priority arrival (e.g: a process entering Q_1) immediately preempts a process running in a lower-priority queue (Q_2 or Q_3)

Execution Trace:

T=0 to T=4: P_1 arrives and enters Q_1. It runs for its full $q=4$ quantum. P_2 arrives at T=2 but must wait because it's the same priority as P_1. At T=4, P_1 is demoted to Q_2.

T=4 to T=8: P_2 starts in Q_1, it runs for its quantum $q = \cancel{8}\ 4$ and

Date:_____ Day:_____

is demoted to Q_2 at $T=8$.

$T=8$ to $T=10$: Q_1 is empty. P_1 (now in Q_2) begins its second execution.

$T=10$ to $T=14$: P_3 arrives in Q_1. Because Q_1 has higher priority than Q_2, P_3 preempts P_1 immediately. P_3 runs for its $q=4$ quantum and is demoted to Q_2 at $T=14$.

$T=14$ to $T=15$: Q_1 is empty. P_2 (the next in line for Q_2) runs for 1 unit.

$T=15$ to $T=18$: P_4 arrives in Q_1. It preempts P_2 immediately. However, at $T=15$, P_2 has actually finished its total 5 unit burst (4 in Q_1 + 1 in Q_2) P_4 runs for its 3-unit burst and completes.

$T=18$ to $T=26$: Q_1 is empty. P_1 resumes in Q_2. It runs for its $q=8$ quantum. By $T=26$, P_1 has used 14 total units $(4+2+8)$. It is demoted to Q_3.

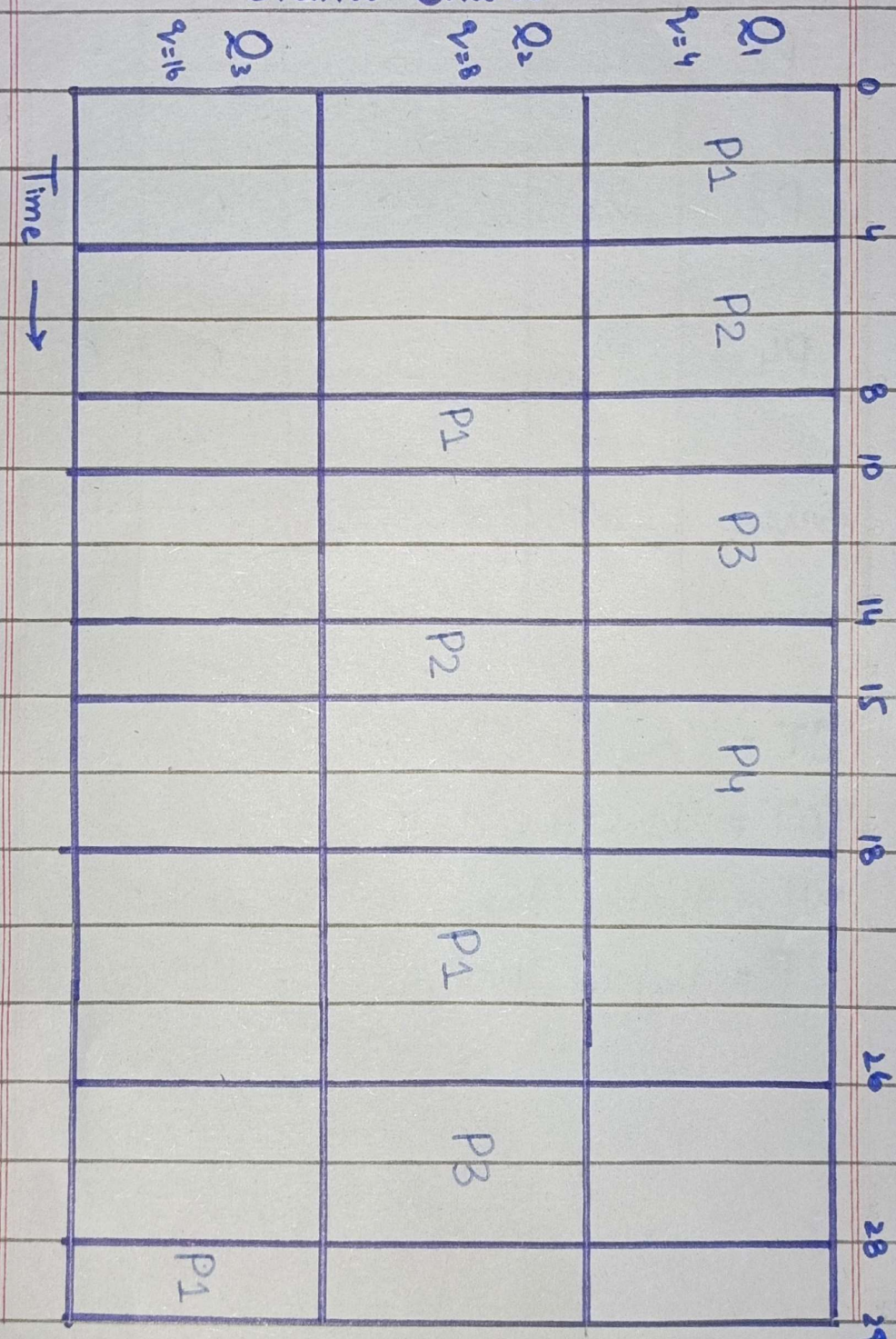
CamScanner

Date: Day:

T = 26 to T = 28: P_3 (in Q_2) runs its remaining 2 units and completes.

T = 28 to T = 29: P_1 (now in Q_3) runs its final 1 unit and completes.

Gantt Chart



Date: Day:

Final Table

Process	CT	TAT (CT-AT)	RT	WT
P1	29	29	0	14
P2	15	13	2	8
P3	28	18	0	12
P4	18	3	0	0
Average	22.5	15.75	0.5	8.5

CT = Completion Time

TAT = Turn Around Time

WT = Waiting Time

RT = Response Time

CamScanner